Prabhavit Gupta



AI Platform Engineer: Agent Runtimes, MCP Tooling, Orchestration & Production Systems

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Summary

AI Platform and Software Engineer with 3 years of hands-on production experience architecting agent runtimes, Model Context Protocol (MCP) integrations, and developer-facing tool layers. Proven record designing and scaling production-grade generative AI applications, grounded retrieval pipelines, and enterprise automation infrastructure. Core technical foundation spanning Python (FastAPI, asyncio), TypeScript/JavaScript (Node.js, Next.js), PostgreSQL, and cloud deployments, with deep practical specialization in autonomous agent architectures: multi-step orchestration, tool and function calling, state management, schema-enforced validation, and automated evaluation loops over live operational logs. Pragmatic builder who operates daily with AI-native coding workflows (Claude Code, Cursor) and designs hardened platform foundations that make it trivial for internal engineering teams to ship reliable agents in days. Holder of EU Long-Term Residence Permit (permanent unrestricted EU work authorization without visa sponsorship); based in Venice, Italy and ready for immediate relocation to Prague.

Technical Expertise

- **Agent Platforms & Runtimes:** Multi-step agent orchestration, LangGraph, tool and function calling, state and memory management, Model Context Protocol (MCP) server development, execution sandboxes, guardrails, fallback escalation.
- **Production LLMOps & Evals:** Automated evaluation pipelines against live production transcripts, hallucination prevention, latency profiling, token economics, context window optimization, semantic caching, trace observability.
- **Retrieval & Data Systems:** Retrieval-Augmented Generation (RAG), vector embeddings, hybrid keyword/semantic search, PostgreSQL, Supabase, relational schema modeling, schema-enforced JSON validation.
- **Full-Stack & Backend Engineering:** Python (FastAPI, backend services, asyncio), TypeScript / JavaScript (Node.js, Next.js, React), RESTful APIs, webhook synchronization, automated contract and regression testing (Playwright).
- **Platform Infrastructure & DevOps:** Docker containerization, AWS/GCP cloud environments, CI/CD pipelines, Git, workflow orchestration (n8n), security best practices (prompt injection defense, audit logging, least-privilege data access).
- **Languages & Authorization:** English (C1 Professional), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (permanent unrestricted EU work authorization, no visa sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer (Full-time, Freelance) *London, UK (remote)*

- Architected, deployed, and maintained the production AI customer agent runtime on a live platform serving 4,500+ active users, automating customer inquiries across pricing, venue inventory, and bookings.
- Designed and operated the tool and connector layer grounding agent responses against live operational databases and partner APIs (OfficeRnD, Nexudus) using strict JSON tool schemas and fallback escalation logic.
- Built automated evaluation loops analyzing real customer conversation transcripts rather than synthetic benchmarks, systematically detecting failure patterns and eliminating hallucinated answers.
- Profiled latency and API token consumption across model versions, optimizing prompt templates and context window utilization to reduce inference costs while maintaining sub-second response times.
- Deployed automated end-to-end regression suites using Playwright, verifying API contracts and continuous deployment stability across complex booking journeys.
- Spearheaded platform migration from legacy architecture to a modern Next.js and backend service stack, supporting platform growth from 2,000 to 4,500+ users without operational degradation.
- Built automated data ingestion and transformation pipelines parsing and validating unstructured partner venue listings, expanding platform inventory to over 900 locations.

Urlink **Apr 2025 - Jul 2025**
Project Manager & AI Assisted Developer (Freelance) *Barcelona, Spain (remote)*

- Architected core web application backend workflows, defining clear API contracts, relational schemas, and data transformation logic for user-facing features.

- Implemented automated validation and synchronization pipelines connecting third-party services with PostgreSQL, preventing malformed inputs and ensuring relational data integrity.
- Integrated automated CI/CD workflows and regression checks to accelerate feature delivery across distributed development environments.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Solutions & Automation Engineer *Venice, Italy*

- Designed and delivered bespoke AI automation workflows, web extraction engines, and API integrations for European SME clients using Python, TypeScript, and Playwright.
- Translated manual, fragmented business operations into automated software solutions with structured logging, retry queues, and reliable recovery logic.
- Engineered webhook listeners and real-time synchronization pipelines across CRM platforms, operational tools, and relational datastores.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous grounding in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.